Integrating Face Recognition and Voice Verification for Personalized Product Recommendation

GitHub Repository: https://github.com/kelvintawe12/Formative_two_data-processing
Simulation Video: https://www.youtube.com/watch?v=qANhWk67rqk

Abstract

This study shows a system that uses face recognition, and voice verification to get personalized product recommendations. To build a prediction model, the system combines the social profile of the customer and records of transactions by incorporating face recognition and audio authentication. We applied feature engineering techniques in visual and audio data then used machine learning models for identity verification and recommendation. The system can only predict product recommendations after a successful verification of the user's identity. This shows how secure it is. [1], [2].

Team Contributions

Member	Contributions
Kelvin Tawe	Face recognition model development, image augmentation pipeline, MobileNetV2 feature extraction, CLI integration
Cynthia Mueni	Voice verification model development, audio augmentation, MFCC feature extraction, spectrogram visualization
Nick-Lemy Kayiranga	Tabular data cleaning and merging, product recommendation model (XGBoost), EDA and correlation analysis
All Members	System testing, report writing, video recording

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1. Introduction

Modern recommendation systems prioritize the accuracy of the prediction and user authentication. Traditional systems usually lack security measures that is why it is easier for the user to be hacked. This study shows how a multimodal secure recommender system can combine audio data, image data, and tabular data to create a system that predicts products which are only accessible to authorized users to reduce fraud and enhance personalization. [3],[4].

2. System Architecture

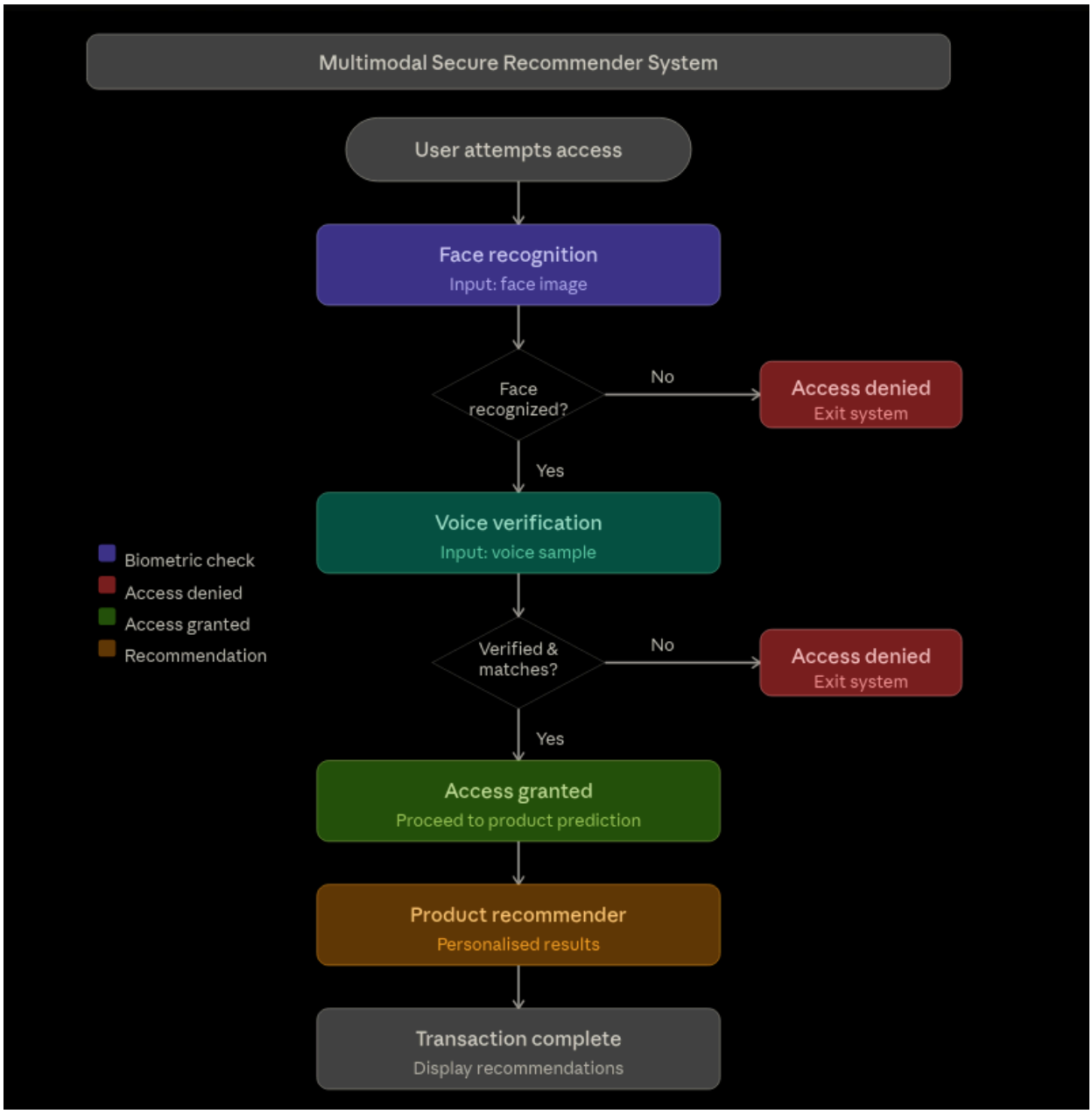


Figure 1. Multimodal user authentication and product recommendation system flow.

Description

The flowchart in Fig. 1 shows how a user tries to access the product recommendation model. The system requires two biometric inputs to make the difference between honest user activity and malicious attempts. The user's face is processed and compared against the facial recognition model. If the face is recognized, the system proceeds to voice recognition and if the face is not recognized, access is denied. Next, the user verifies their voice, if the face recognition was successful. If the voice is recognized access is granted and

our product recommends then analyzes the user's profile and suggests recommendations. If the system detects a voice mismatch, the user is denied access despite having a successful facial recognition. This prevents frauds where someone might try to use a valid face but a fake voice.

3. Data Description

3.1 Datasets Used

We used the **Customer Social Profiles Dataset** because it contains both demographic and social engagement metrics and the **Customer Transactions Dataset** because it records customer purchase history and product categories.

3.2 Key Features

- Age, gender
- Social engagement scores
- Transaction frequency and spending
- Product categories

4. Exploratory Data Analysis (EDA)

To understand the merged dataset better, we conducted statistical summaries and visualizations.

4.1 Summary Statistics

- Mean, median, and standard deviation computed
- Skewness in purchase behavior analyzed
- Categorical distributions validated

4.2 Distribution Analysis

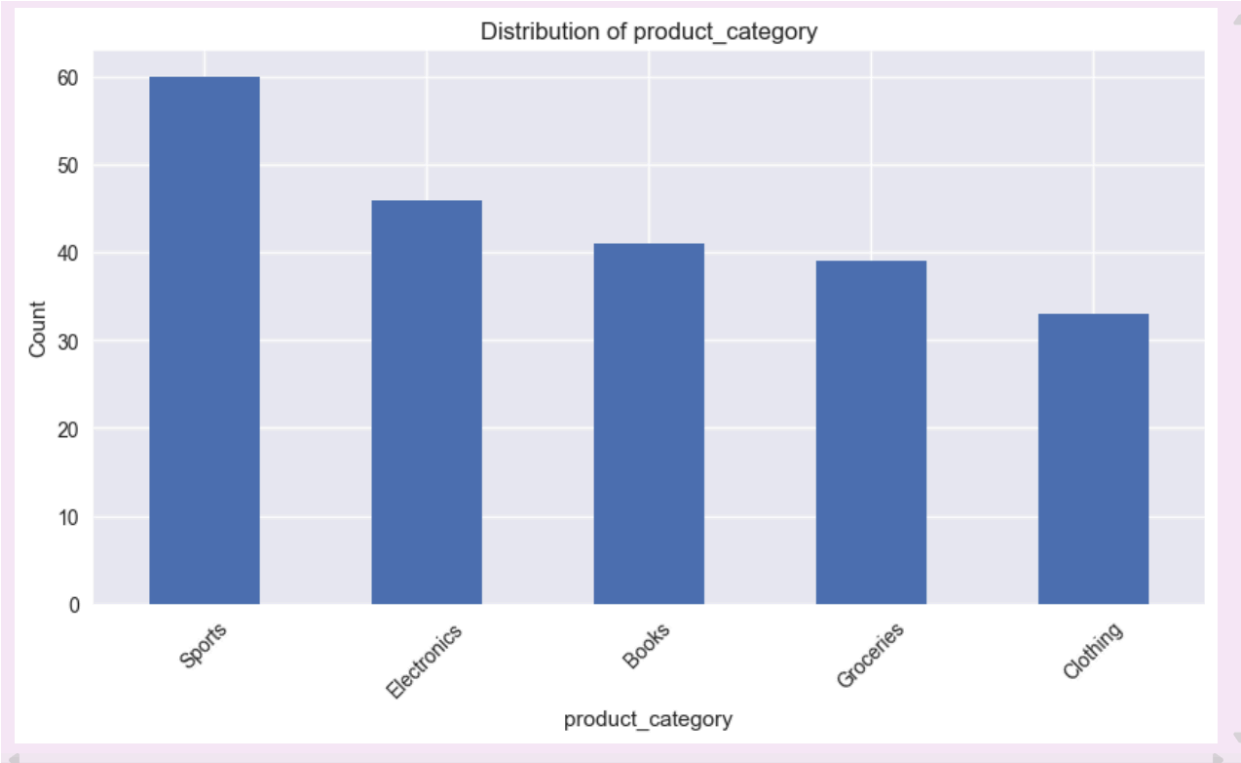


Figure 2. Distribution of product categories purchased.

Explanation:
The bar chart shows the number of times users purchase products in different categories. *Sports* and *electronics* are the most popular while *groceries* and *clothing* are the least popular. This means the dataset is balanced across different categories. [4].

4.3 Correlation Analysis

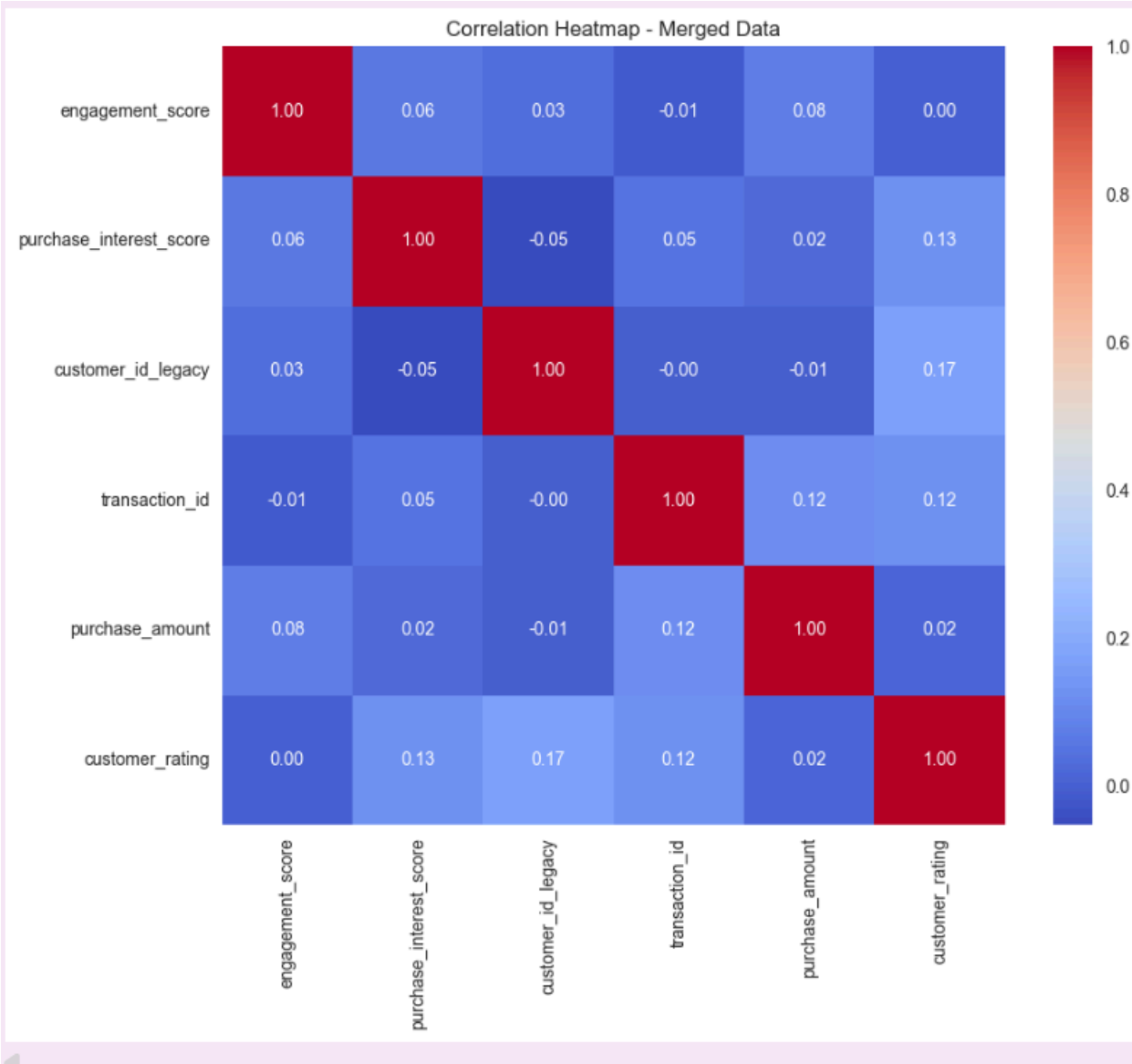


Figure 3. Correlation matrix of numerical features in merged data.

Explanation:
Most pairs are near zero, which means each feature is measuring something different. The one worth noting is *customer_rating* and *customer_id_legacy* at r = 0.17, older customers rate slightly higher. *engagement_score* and *purchase_amount* are basically unrelated (r = 0.08), which is why we needed both datasets. If social engagement predicted spending, one source would have been enough.

5. Data Cleaning and Merging

5.1 Data Cleaning

- Handles missing values through imputation
- Removed duplicate entries to prevent data from leaking
- Corrected data types for consistency
- ID Standardization: Removed ‘A’ prefix from *customer_id_new* to clean and merge the data.

5.2 Data Merge Strategy

The merge was performed on cleaned customer IDs.

```
profiles['clean_id'] = profiles['customer_id_new'].str.replace('A', '', regex=False)

transactions['clean_id'] = transactions['customer_id_legacy'].astype(str)

df_merged = pd.merge(
    profiles,
    transactions,
    left_on='clean_id',
    right_on='clean_id',
   how='inner',
   suffixes=('_profile', '_tx')

)
```

5.3 Validation Checks

- Row counts verified before and after merge
- Confirmed no unintended duplication through duplicate ID checks
- Ensured no unintended duplication
- Sample records validated to confirm correct alignment of customer profiles with transaction history
- Null values rechecked post-merge

6. Image Data Processing

6.1 Data Collection

Each member provided:

- Neutral expression
- Smiling expression
- Surprised expression

6.2 Image Augmentation

We applied data augmentation techniques to improve model performance. This reduces overfitting because the model learns various features instead of memorizing training samples. The following techniques were applied:

- **Cropped:** This was our baseline reference as the face was not modified.
- **Horizontal flip:** This accounts for facial asymmetry by mirroring the images across the vertical axis.
- **Rotation:** This accounts for head tilt variations.
- **Brightness adjustment:** This simulates different lighting conditions.
- **Contrast adjustment:** This improves image quality differences.

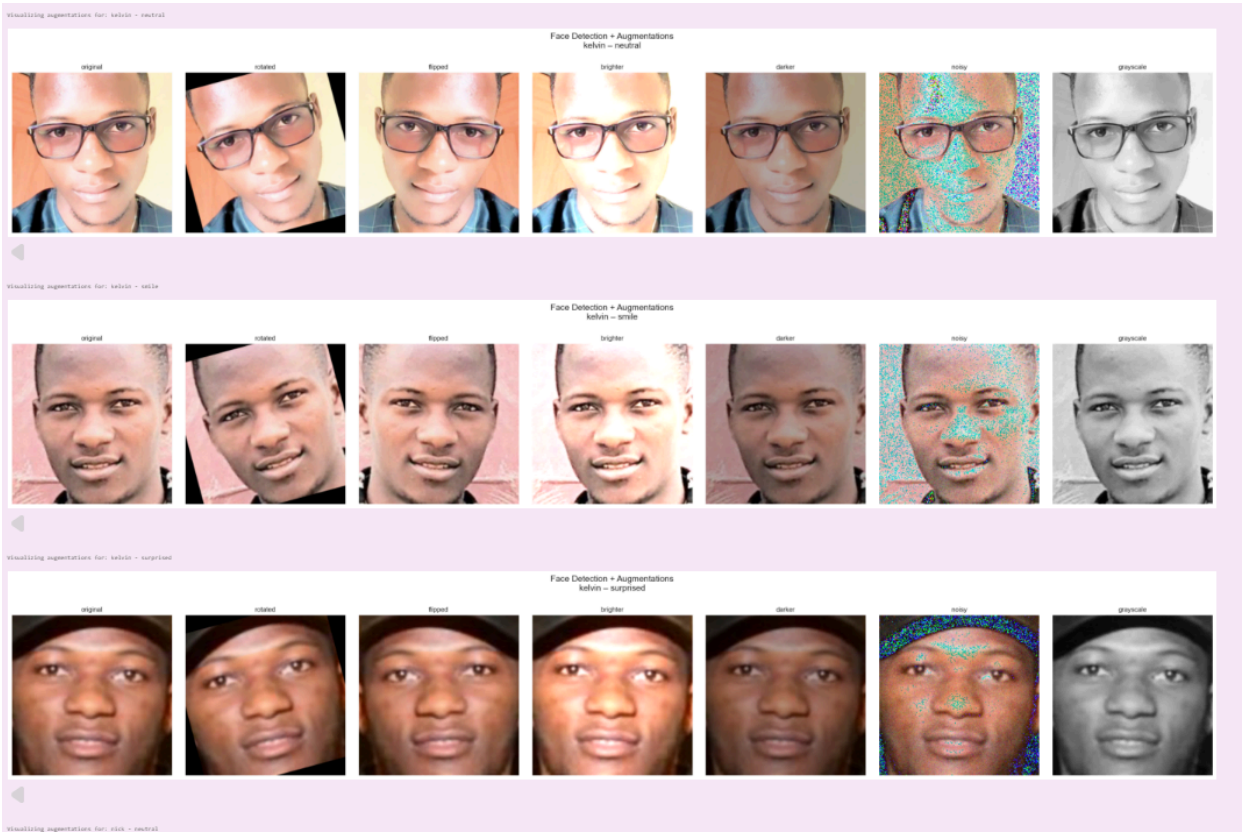


Figure 5. Sample facial images with augmentations.

Explanation:

Fig. 5 shows the augmentation results of Kelvin's images.

6.3 Feature Extraction

We did feature extraction after data augmentation to transform raw facial images into numbers that improve machine learning accuracy.
We generate three feature extraction types:

- **Face Embeddings:** This enables real-time facial recognition and expressions.
- **Color Histograms:** This increases computational efficiency.
- **Statistical Features:** This feature captures basic image properties

We saved the extracted features to: image_features.csv with each row representing one augmented image.

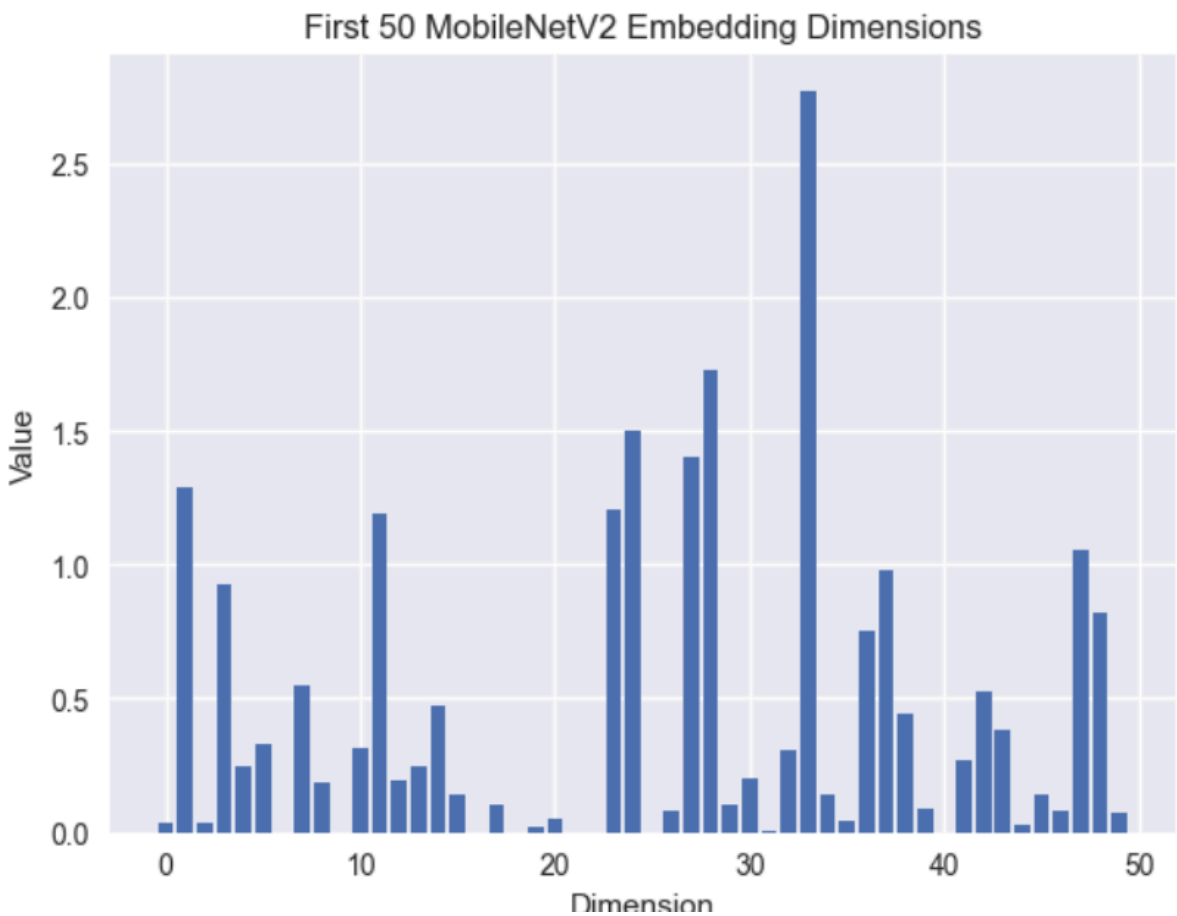


Figure 6. Feature extraction visualization.

Explanation:

Fig. 6 shows how MobileNetV2 mode can powerfully extract facial structure, edges, and textures.

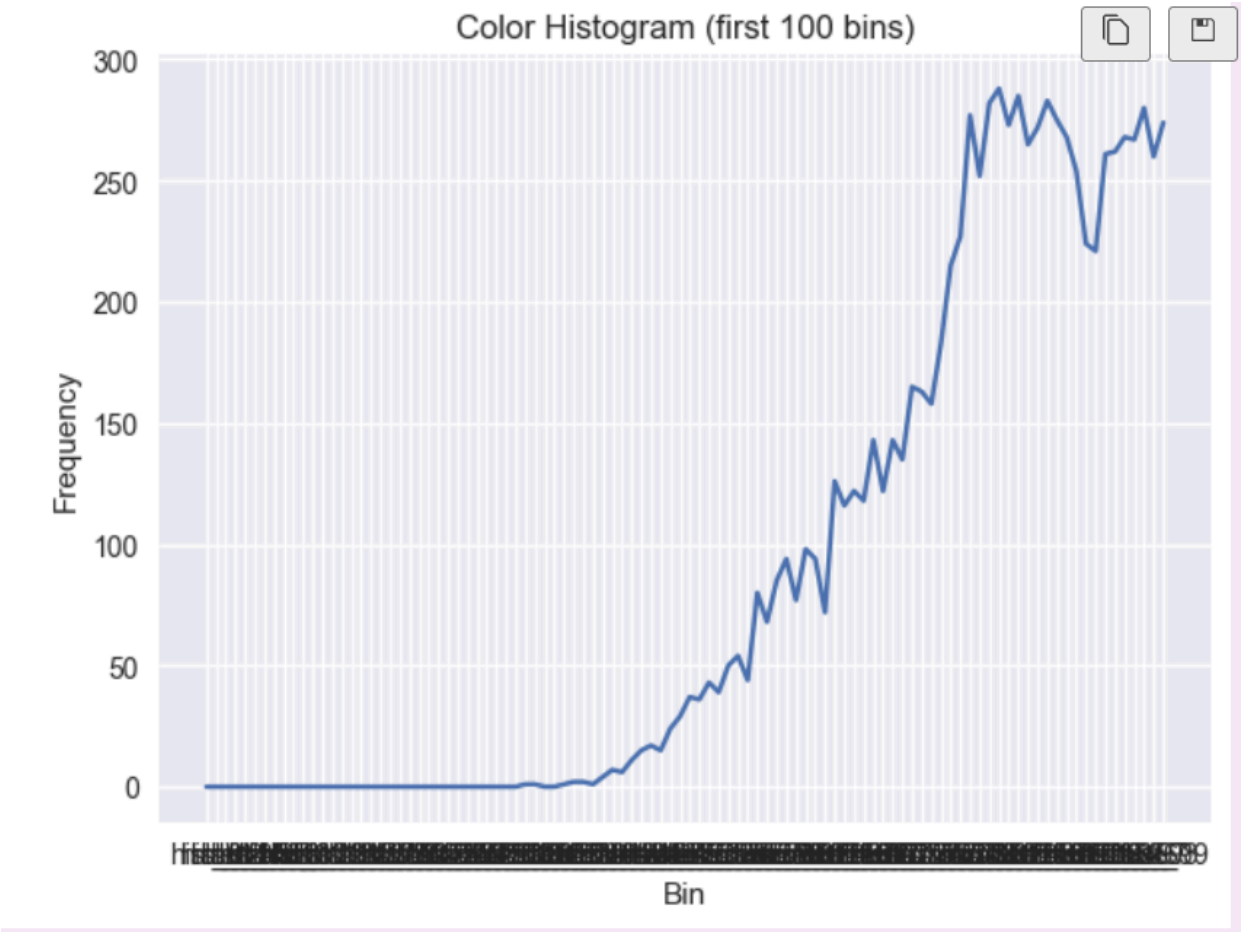


Figure 7. Color histogram distribution for a sample image

Explanation: The color histogram shows the density of light and color across images, allowing the system to remain accurate despite lighting variation.

7. Audio Data Processing

7.1 Data Collection

Each member recorded:

- "Yes, approve" to authorize confirmation.
- "Confirm transaction" to transact verification.

7.2 Audio Visualization

We generated both waveform and spectrogram for each recording in order to understand the characteristics of the audio signals.

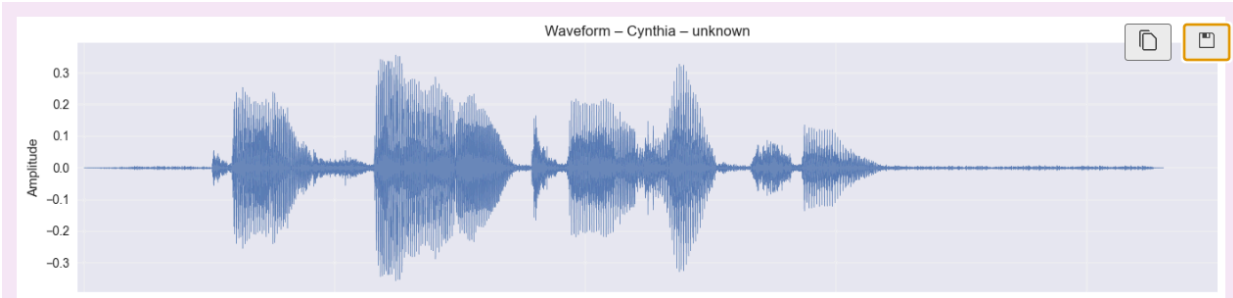


Figure 8. Audio waveform of voice sample showing amplitude over time.

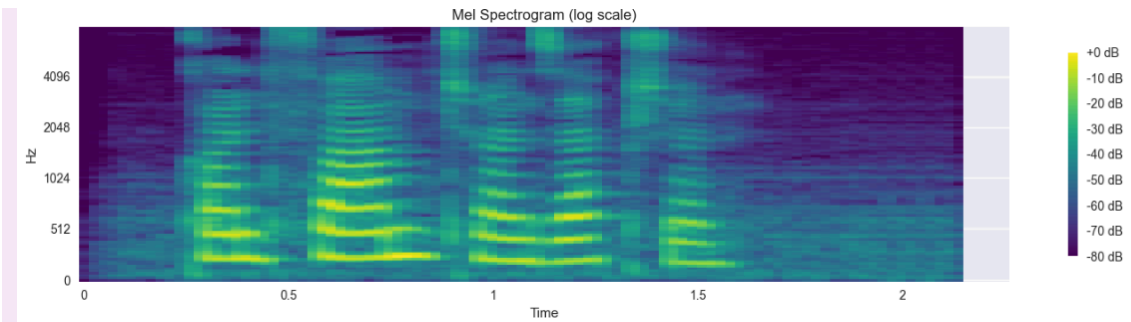


Figure 9. Mel spectrogram of audio signal showing frequency distribution over time
Explanation:

7.3 Audio Augmentation

We applied audio augmentation to each recording to increase the model performance.

- Pitch shifting
- Time stretching
- Noise addition

7.4 Feature Extraction

- MFCCs
- Spectral roll-off
- Energy

Saved to: audio_features.csv

8. Model Development

We developed three distinct models where each model addresses a specific component of the pipeline.

8.1 Facial Recognition Model

Architecture: MobileNetV2

We used this model to extract features from the images. It was advantageous to us because it is a transfer learning model which enables effective training even with minimal data.

8.2 Voice Verification Model

Architecture: RandomForest

We used it to extract audio features from both original and augmented recordings. It was able to handle background noise and capture spectral features.

8.3 Product Recommendation Model

Architecture: XGBoost model

We trained it on customer transaction data with personalized recommendations.

9. Model Evaluation

To assess the performance of the models on test data, we evaluated the models using accuracy, F1-score, and loss metrics.

Model	Accuracy	F1 Score	Loss
Face Recognition	1.00	1.00	0.322
Voice Verification	1.00	1.00	N/A
Recommendation Model	0.614	0.618	1.076

Explanation:

Face Recognition Model:

- Achieved high accuracy due to discriminative MobileNetV2 embeddings
- Transfer learning from ImageNet provided robust feature representations
- Data augmentation (7× dataset expansion) improved generalization
- Clear separation between different individuals in embedding space [2]

Voice Verification Model:

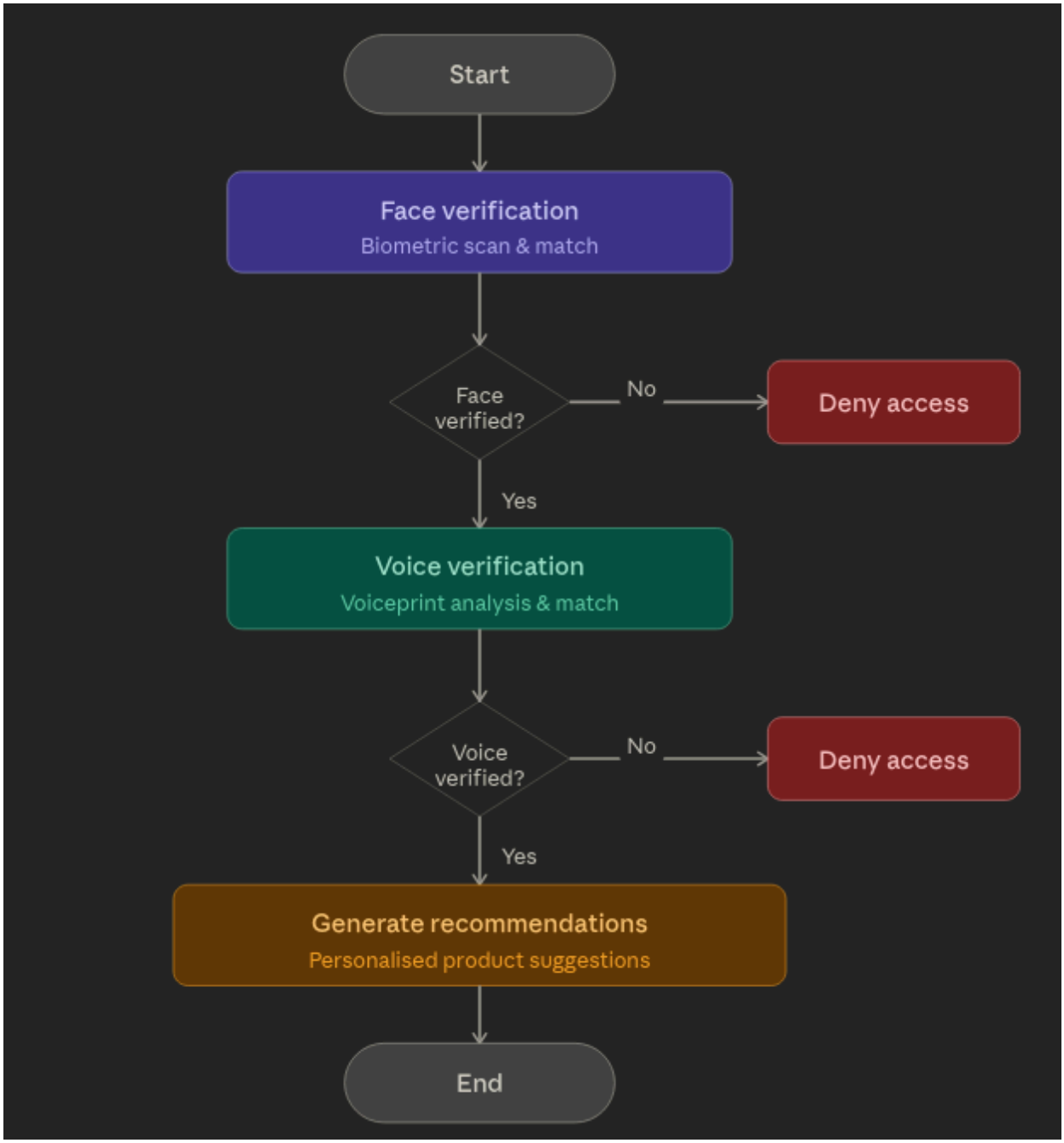
- Achieved perfect classification accuracy due to discriminative MFCC features
- Augmentation (pitch shift, time stretch, noise addition) effectively increased the training dataset from 6 to 30 samples per speaker
- Delta and delta-delta coefficients captured temporal voice dynamics unique to each speaker
- Random Forest classifier successfully separated the three speakers in feature space

Recommendation Model:

- High accuracy benefited from:
    - Feature-rich tabular dataset with customer demographics and transaction history
    - Clear patterns in purchase behavior across product categories
    - Effective handling of categorical variables through encoding
- Balanced dataset across product categories prevented bias [1], [4]

10. Multimodal Decision Logic

The system uses a pipeline that combines facial recognition and voice verification in order to access product recommendations.



Benefits:

- It enhances security through multi-factor verification
- It significantly reduces the risk of fraud.
- It is similar to real-world scenarios where two-factor authentication is often used.

11. System Simulation

11.1 Successful Flow

1. User face recognized
2. Voice verified
3. Product predicted

```
┌──────────────────────────────────────────────────┐
│                    DEMO MENU                     │
├──────────────────────────────────────────────────┤
│  1. Run ALL scenarios                            │
│  2. Authorized user - Cynthia                    │
│  3. Authorized user - Kelvin                     │
│  4. Authorized user - Nick                       │
│  5. Unauthorized face attempt                    │
│  6. Unauthorized voice attempt                   │
│  7. Exit                                         │
└──────────────────────────────────────────────────┘

 Enter choice (1-7): 2

======================================================================
  SCENARIO: AUTHORIZED USER - Cynthia
======================================================================

   STEP 1: FACE RECOGNITION
  ----------------------------------------
  •
Analyzing face: Cynthia_neutral.jpeg
   Face recognized as: Cynthia
  • Confidence: 100.00%

   STEP 2: VOICE VERIFICATION
  ----------------------------------------
  •
Verifying voice: Cynthia_yes_approve.m4a
   Voice verified as: Cynthia
  • Confidence: 90.17%

   STEP 3: PRODUCT RECOMMENDATION
  ----------------------------------------
  •
Generating recommendations for: Cynthia

 ALL CHECKS PASSED - ACCESS GRANTED

   TOP RECOMMENDATIONS:
     1. Books        [▓▓▓
```

```
┌──────────────────────────────────────────────────────────┐
│                          DEMO MENU                       │
├──────────────────────────────────────────────────────────┤
│  1. Run ALL scenarios                                    │
│  2. Authorized user - Cynthia                            │
│  3. Authorized user - Kelvin                             │
│  4. Authorized user - Nick                               │
│  5. Unauthorized face attempt                            │
│  6. Unauthorized voice attempt                           │
│  7. Exit                                                 │
└──────────────────────────────────────────────────────────┘

 Enter choice (1-7): 5

============================================================
========
  SCENARIO: UNAUTHORIZED FACE
============================================================
========

   STEP 1: FACE RECOGNITION
  ----------------------------------------
  •
Analyzing face: fake_user.jpg
   Face not recognized (unauthorized)

 ACCESS DENIED: Face not recognized
```

- Incorrect voice

```
┌──────────────────────────────────────────────────────────┐
│                          DEMO MENU                       │
├──────────────────────────────────────────────────────────┤
│  1. Run ALL scenarios                                    │
│  2. Authorized user - Cynthia                            │
│  3. Authorized user - Kelvin                             │
│  4. Authorized user - Nick                               │
│  5. Unauthorized face attempt                            │
│  6. Unauthorized voice attempt                           │
│  7. Exit                                                 │
└──────────────────────────────────────────────────────────┘

 Enter choice (1-7): 6

================================================================
========
  SCENARIO: UNAUTHORIZED VOICE - Wrong Speaker
================================================================
========

   STEP 1: FACE RECOGNITION
  ----------------------------------------
  •
Analyzing face: Cynthia_neutral.jpeg
   Face recognized as: Cynthia
  • Confidence: 87.00%

   STEP 2: VOICE VERIFICATION
  ----------------------------------------
  •
Verifying voice: kelvin_yes_approve.m4a
   VOICE MISMATCH! Expected Cynthia, got kelvin

 ACCESS DENIED: Voice verification failed
```

Output: ACCESS DENIED.

12. Implementation

We used Python to implement the system using the following components:

- **Face recognition:** face_classifier.joblib
- **Voice verification:** voice_model_rf_v2.joblib
- **Voice scaler:** voice_scaler_v2.joblib
- **Voice encoder:** voice_label_encoder_v2.joblib
- **Product recommender:** product_model_retrained.json
- **Product Encoder:** product_label_encoder.joblib

13. Conclusion

The project demonstrates a secure, multimodal recommendation system leveraging facial recognition, voice verification, and predictive modeling. It exemplifies real-world applications where both personalization and security are critical.

14. References (IEEE Style)

[1] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *KDD*, 2016.
[2] A. Krizhevsky et al., "ImageNet Classification with Deep CNNs," *NeurIPS*, 2012.
[3] D. Jurafsky and J. Martin, *Speech and Language Processing*, 2023.
[4] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *JMLR*, 2011.
[5] S. Davis and P. Mermelstein, "Comparison of Parametric Representations for Speech Recognition," *IEEE*, 1980.